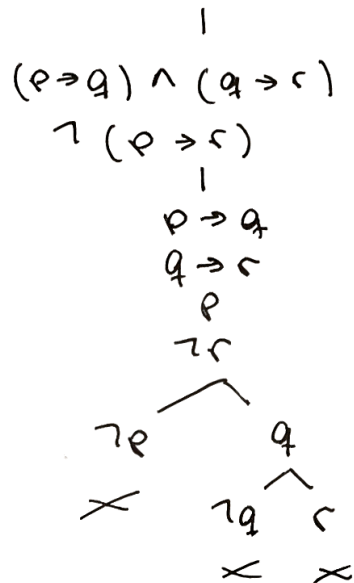


Boletín de Lógica e conjuntos

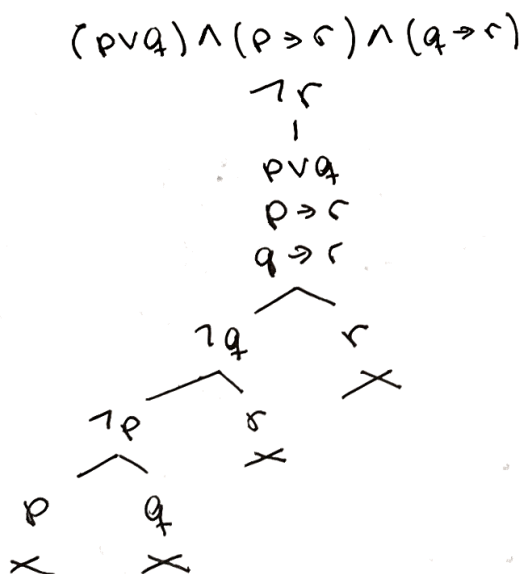
1. a. $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r) \quad T$

$\perp \quad \neg [[p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)]$



b. $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r \quad T$

$\perp \quad \neg [[p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r]$



2 $[\neg q \wedge (p \Rightarrow q)] \rightarrow \neg q$

T

$\perp \neg [\neg q \wedge (p \Rightarrow q)] \Rightarrow \neg q$

$$\begin{array}{c} \neg q \wedge (p \Rightarrow q) \\ \neg \neg q \\ \neg q \\ q \\ \times \end{array}$$

3 $[(p \Rightarrow q) \rightarrow r]$

$$\begin{array}{c} \neg(p \Rightarrow q) \quad r \\ \neg p \quad q \\ p \quad \neg q \end{array}$$

$[(p \Rightarrow (q \Rightarrow r))]$

$$\begin{array}{c} \neg p \quad q \Rightarrow r \\ \neg q \quad r \end{array}$$

Na primeira, com q falso e r verdade, p é verdade. Mentres que na segunda, com q falso e r verdade, p é falso.

4 a $\frac{\neg p \rightarrow (q \rightarrow r)}{\neg(\neg p) \vee (q \rightarrow r)} \frac{\neg(\neg p) \vee (q \rightarrow r)}{p \vee \neg q \vee r}$

b $\frac{\neg(q \Rightarrow p) \Rightarrow r}{\neg\neg(q \Rightarrow p) \vee r} \frac{\neg\neg(q \Rightarrow p) \vee r}{\neg q \vee p \vee r}$

c $\frac{(q \Rightarrow p) \vee (\neg p \Rightarrow r)}{\neg q \vee p \vee \neg p \vee r} \frac{\neg q \vee p \vee \neg p \vee r}{\neg q \vee p \vee r}$

5 a $\forall x (Q(x, \top))$

b $\forall x \exists y (Q(x, y))$

c $\forall x \exists y (\neg Q(y, x))$

d $\exists x (\neg Q(\perp, x))$

e $\forall x \exists y (\neg Q(x, y))$

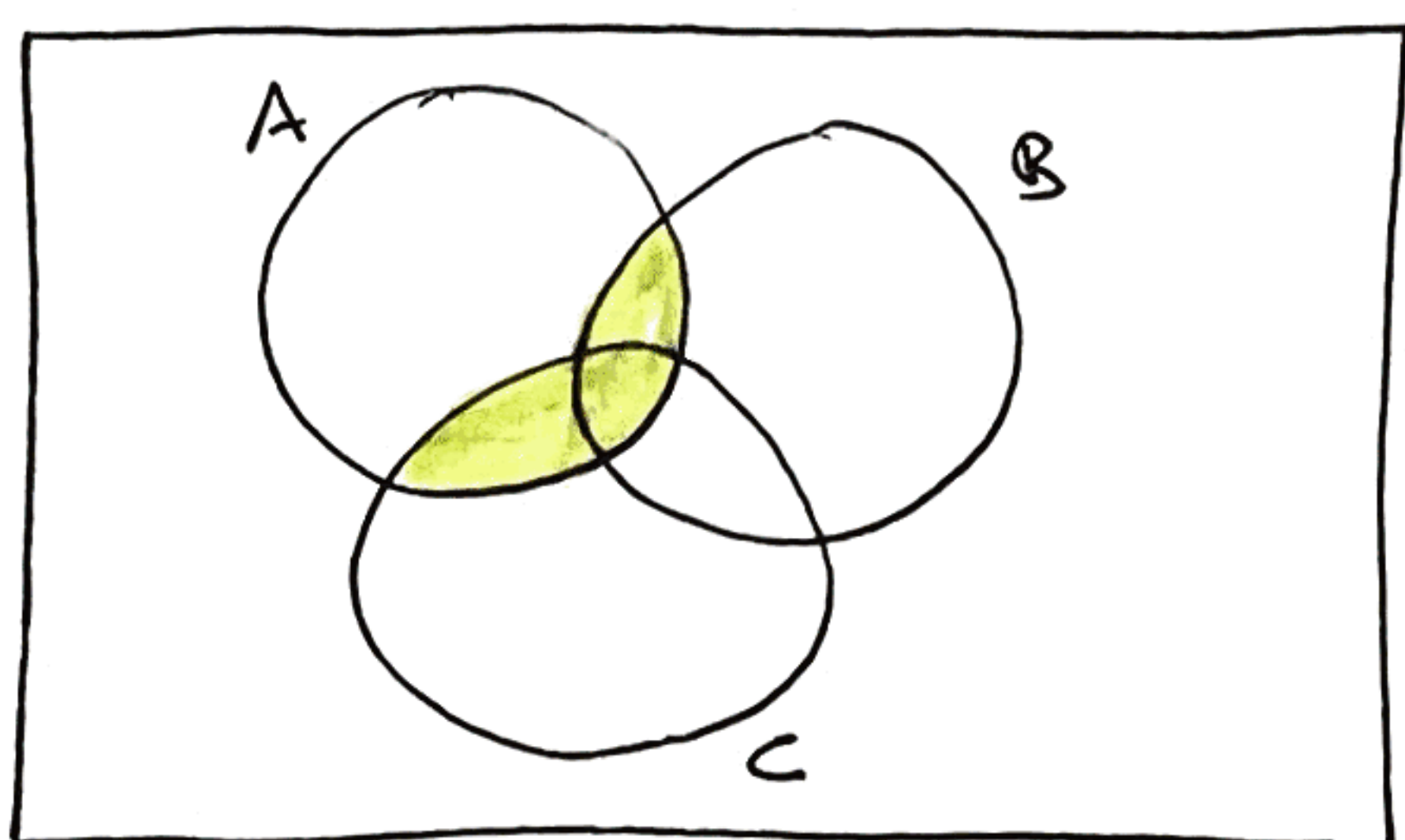
f $\forall x \exists y_1, \exists y_2 (Q(x, y_1) \wedge Q(x, y_2) \wedge \neg(y_1, y_2))$

g $\exists x \exists y (Q(x, y) \wedge \neg(x, y))$

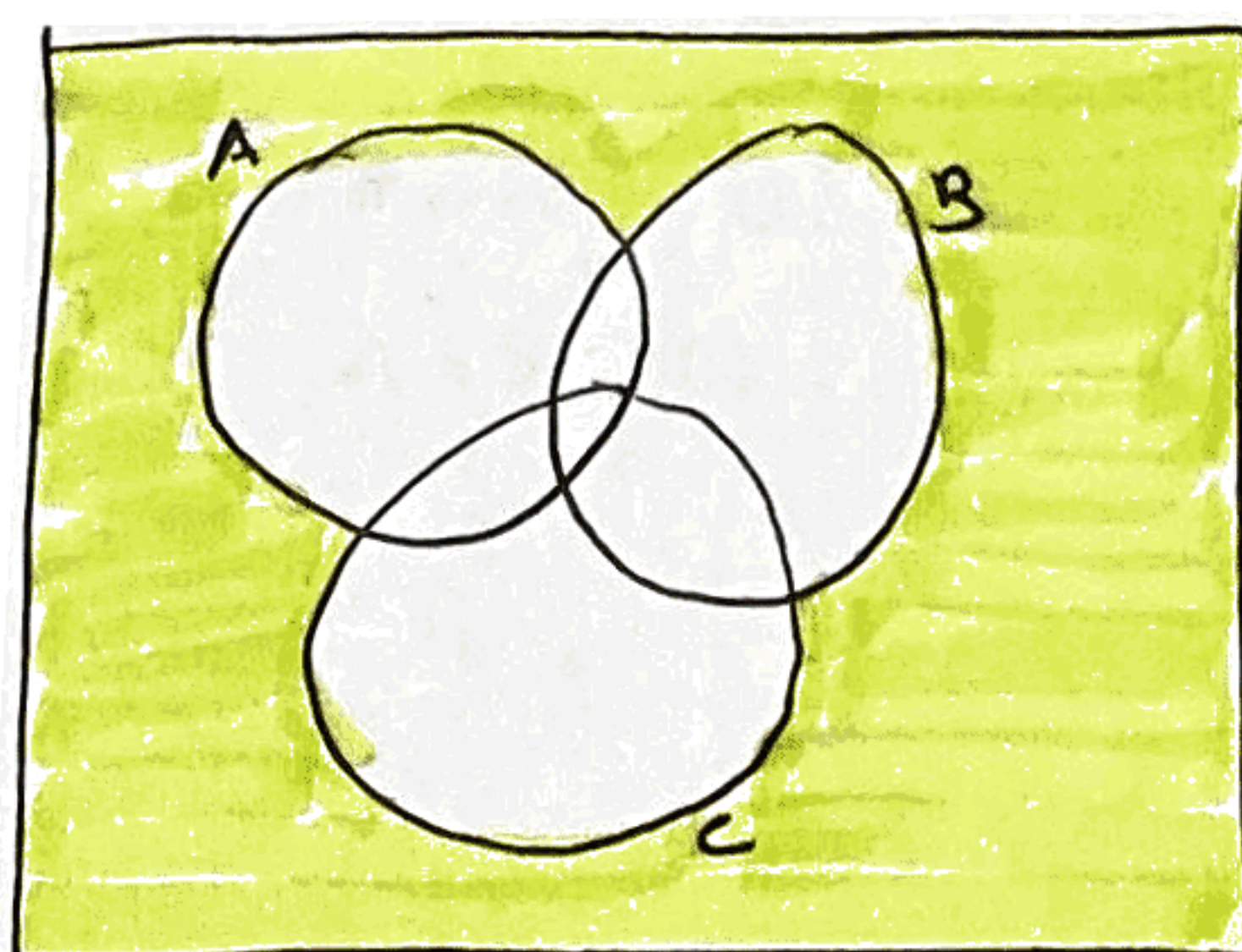
- 6
- a $\emptyset \in \{\emptyset\} \quad V$
 - b $\{\{\emptyset\}\} \in \{\emptyset, \{\emptyset\}\} \quad F$
 - c $\{\emptyset\} \in \{\emptyset\} \quad F$
 - d $\{\emptyset\} \in \{\{\emptyset\}\} \quad V$
 - e $\{\emptyset\} \subseteq \{\emptyset, \{\emptyset\}\} \quad V$

- f $\{\{\emptyset\}\} \subseteq \{\emptyset, \{\emptyset\}\} \quad V$
- g $\{\{\emptyset\}\} \subset \{\{\emptyset\}, \{\emptyset\}\} \quad F$
- h $\emptyset \notin \{\{\{\emptyset\}\}, \{\emptyset\}\} \quad V$
- i $\{\emptyset\} \subseteq \{\{\emptyset\}, \{\emptyset\}\} \quad F$

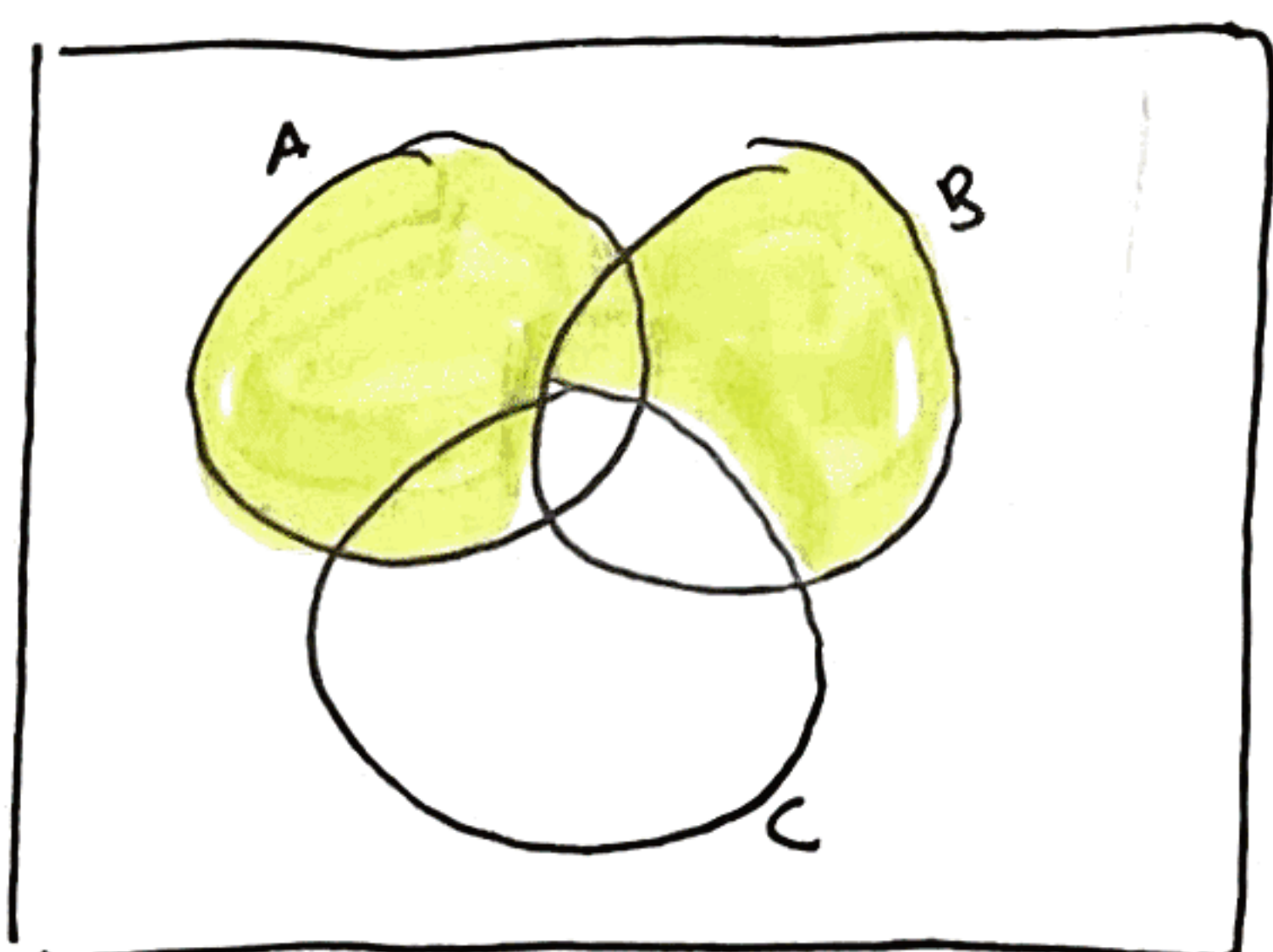
7 a $A \cap (B \cup C)$



b $\bar{A} \cap \bar{B} \cap \bar{C}$



c $(A \setminus B) \cup (A \setminus C) \cup (B \setminus C)$



8 a $A \cup B = A$

- $A \cup B \subseteq A \Rightarrow x \in A \cup B \rightarrow x \in A \Rightarrow \begin{cases} x \in A \\ \text{or} \\ x \in B \end{cases} \rightarrow \underline{x \in A}$
- $A \subseteq A \cup B$ (sempre se ungle)

$B \subseteq A$

b $A \cap B = A$

- $A \cap B \subseteq A$ (sempre se ungle)
- $A \subseteq A \cap B \Rightarrow x \in A \rightarrow x \in A \cap B \Rightarrow \underline{x \in A} \rightarrow \begin{cases} x \in A \\ \text{and} \\ x \in B \end{cases}$

$A \subseteq B$

c $A \setminus B = A$

- $A \setminus B \subseteq A \Rightarrow$ sempre se ungle
- $A \subseteq A \setminus B \Rightarrow x \in A \rightarrow x \in A \setminus B \quad \left\{ \begin{array}{l} x \in A \Rightarrow \text{sempre} \\ x \notin B \Rightarrow A \cap B = \emptyset \end{array} \right.$

d $A \cap B = B \cap A$

- $A \cap B \subseteq B \cap A \Rightarrow x \in A \cap B \quad \left\{ \begin{array}{l} x \in A \rightarrow x \in B \cap A \\ x \in B \rightarrow x \in B \cap A \end{array} \right. \quad \left\{ \begin{array}{l} x \in B \Rightarrow A \subseteq B \\ x \in A \Rightarrow \text{sempre} \end{array} \right. \quad \left. \vphantom{\left\{ \begin{array}{l} x \in A \rightarrow x \in B \cap A \\ x \in B \rightarrow x \in B \cap A \end{array} \right.}} \right\} A=B$
- $B \cap A \subseteq A \cap B \Rightarrow x \in B \cap A \quad \left\{ \begin{array}{l} x \in B \rightarrow x \in A \cap B \\ x \in A \rightarrow x \in A \cap B \end{array} \right. \quad \left\{ \begin{array}{l} x \in A \Rightarrow B \subseteq A \\ x \in B \Rightarrow \text{sempre} \end{array} \right. \quad \left. \vphantom{\left\{ \begin{array}{l} x \in B \rightarrow x \in A \cap B \\ x \in A \rightarrow x \in A \cap B \end{array} \right.}} \right\} A=B$

e $A \setminus B = B \setminus A$

- $A \setminus B \subseteq B \setminus A \Rightarrow x \in A \setminus B \quad \left\{ \begin{array}{l} x \in A \rightarrow x \in B \setminus A \\ x \notin B \rightarrow x \in B \setminus A \end{array} \right. \quad \left\{ \begin{array}{l} x \in B \Rightarrow A \subseteq B \\ x \notin A \Rightarrow A = \emptyset \end{array} \right.$
- $B \setminus A \subseteq A \setminus B \Rightarrow x \in B \setminus A \quad \left\{ \begin{array}{l} x \in B \rightarrow x \in A \setminus B \\ x \notin A \rightarrow x \in A \setminus B \end{array} \right. \quad \left\{ \begin{array}{l} x \in A \\ x \notin B \Rightarrow B = \emptyset \end{array} \right. \quad \left\{ \begin{array}{l} x \in A \Rightarrow A = \emptyset \\ x \notin B \end{array} \right.$

f $A \cup B = B \cap A$

- $A \cup B \subseteq B \cap A \Rightarrow x \in A \cup B \quad \left\{ \begin{array}{l} x \in A \rightarrow x \in B \cap A \\ x \in B \rightarrow x \in B \cap A \end{array} \right. \quad \left\{ \begin{array}{l} x \in B \Rightarrow A \subseteq B \\ x \in A \rightarrow \text{sempre} \end{array} \right. \quad \left. \vphantom{\left\{ \begin{array}{l} x \in A \rightarrow x \in B \cap A \\ x \in B \rightarrow x \in B \cap A \end{array} \right.}} \right\} A=B$

- $B \cap A \subseteq A \cup B \Rightarrow x \in B \cap A \quad \left\{ \begin{array}{l} x \in B \rightarrow x \in A \cup B \\ x \in A \rightarrow x \in A \cup B \end{array} \right. \quad \left\{ \begin{array}{l} x \in A \\ x \in B \end{array} \right.$